

Composition to Verify Inverses

TEACHER KEY | Algebra II | Unit 1 | Day 13 | TEK A2.2.D

Objective: I can compose two functions and use the results $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$ to prove whether they are inverses of each other.

Vocabulary

Composition of functions — putting one function **inside** another and evaluating: $f(g(x))$, read "f of g of x." Work from the **inside out**.

Part 1 — Warm Up on Composition

Let $f(x) = 3x - 5$ and $g(x) = (x + 5)/3$. Work inside first.

$$g(2) = \mathbf{7/3} \quad f(4) = \mathbf{7} \quad f(g(4)) = f(\mathbf{3}) = \mathbf{4}$$

$$g(f(4)) = g(\mathbf{7}) = \mathbf{4} \quad \text{Notice both came back to } \mathbf{4}, \text{ the number we started with.}$$

Part 2 — Prove It with Algebra

Example 1. $f(x) = 3x - 5$ and $g(x) = (x + 5)/3$

$$f(g(x)) = 3 \cdot (\mathbf{x + 5})/3 - 5 = \mathbf{x + 5 - 5} = \mathbf{x}$$

$$g(f(x)) = (\mathbf{3x - 5} + 5)/3 = \mathbf{3x/3} = \mathbf{x}$$

Both give x , so f and g **are** inverses.

You Try 1. $f(x) = 2x + 3$ and $g(x) = (x - 3)/2$

$$f(g(x)) = \mathbf{x} \quad g(f(x)) = \mathbf{x} \quad \text{Inverses? } \mathbf{yes}$$

You Try 2. $f(x) = x + 6$ and $g(x) = x - 6$

$$f(g(x)) = \mathbf{x} \quad g(f(x)) = \mathbf{x} \quad \text{Inverses? } \mathbf{yes}$$

Part 3 — A Pair That Is NOT Inverse

Test $f(x) = 2x + 1$ and $g(x) = 2x - 1$. Do not assume - compute.

$$f(g(x)) = 2(\mathbf{2x - 1}) + 1 = \mathbf{4x - 1} \quad \text{Does this equal } x? \quad \mathbf{no}$$

So they are **not** inverses. What is the actual inverse of $f(x) = 2x + 1$? $f^{-1}(x) = (\mathbf{x - 1})/2$

Lesson: changing a sign is **not** the same as undoing the function. You must undo every step, in reverse order.

Part 4 — Error Analysis

A student is given $f(x) = 2(x - 3)^2 + 4$ and claims $f^{-1}(x) = \sqrt{x - 3} + 4$.

Test it: is $f(f^{-1}(x)) = x$? **no** So the claim is **wrong**.

Find the mistake. Undo the steps in reverse: subtract 4, divide by 2, take the square root, then add 3.

$$\text{Correct inverse: } f^{-1}(x) = \mathbf{3 \pm \sqrt{(x - 4)/2}}$$

Why does a $\pm$ show up here? **the original is a parabola, so it is not one-to-one**

What restriction would make the inverse a function? **$x \geq 3$ (or $x \leq 3$)**

Part 5 — Self-Inverse Functions

A few functions are their own inverse - composing them with themselves gives x .

$$1. \quad f(x) = x \quad f(f(x)) = \mathbf{x} \quad \text{Self-inverse? } \mathbf{yes}$$

2. $f(x) = -x$ $f(f(x)) = x$ Self-inverse? **yes**

3. $f(x) = 1/x$ $f(f(x)) = x$ Self-inverse? **yes**

What do their graphs share? **each is symmetric about the line $y = x$**

Practice

Decide whether each pair is inverse. Show BOTH compositions - one direction is not proof.

1. $f(x) = 5x$ and $g(x) = x/5$

$f(g(x)) = x$ $g(f(x)) = x$ Inverses? **yes**

2. $f(x) = x - 3$ and $g(x) = 3 - x$

$f(g(x)) = -x$ Inverses? **no** Correct inverse of f : **$x + 3$**

3. $f(x) = (x + 1)/2$ and $g(x) = 2x - 1$

$f(g(x)) = x$ $g(f(x)) = x$ Inverses? **yes**

4. $f(x) = x^2$ and $g(x) = \sqrt{x}$. Test with $x = -4$ as well as $x = 4$.

Does it work for every real x ? **no - it fails for negatives** Restriction needed: **$x \geq 0$**

5. Write the inverse of $f(x) = 7x + 2$, then verify it by composition.

$f^{-1}(x) = \mathbf{(x - 2)/7}$

Exit Ticket

Decide whether $f(x) = 4x - 8$ and $g(x) = (x + 8)/4$ are inverses. Show both compositions.

$f(g(x)) = x$ $g(f(x)) = x$ Inverses? **yes**

Why is checking only one direction not enough? **both compositions must return x for the pair to be inverses**